

Aejandro De La Torre

LIS 440

wk14: EXERCISE -- Pandas Aggregating -- Summarizing and Grouping

12/3/25

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output'
code = root / 'code'

# -----
# Analysis: Totals and Groupings
# -----

# Fetch the data
icecreamDF = pd.read_csv(data_processed / 'icecreamsales100random.csv')
print("\nIce Cream Data: ")
print(icecreamDF)

# Totals for All Transactions
print("Totals for All Transactions:")
totalScoops = icecreamDF['Scoops'].sum()
averageScoopSize = icecreamDF['Scoops'].mean()
print("Total Scoops (All Transactions):", totalScoops)
print("Average Scoop Size (All Transactions):", round(averageScoopSize, 2))

# Conditional Subsetting with Row Values
print("Conditional Subsetting: Flavor")
```

```

# Chocolate
chocolateDF = icecreamDF[icecreamDF['Flavor'] == 'Chocolate']
print("Chocolate Transactions:", chocolateDF.shape[0])
print("Chocolate Total Scoops:", chocolateDF['Scoops'].sum())
print("Chocolate Average Scoops:", round(chocolateDF['Scoops'].mean(), 2))

# Vanilla
vanillaDF = icecreamDF[icecreamDF['Flavor'] == 'Vanilla']
print("Vanilla Transactions:", vanillaDF.shape[0])
print("Vanilla Total Scoops:", vanillaDF['Scoops'].sum())
print("Vanilla Average Scoops:", round(vanillaDF['Scoops'].mean(), 2))

# Blue Moon
blueMoonDF = icecreamDF[icecreamDF['Flavor'] == 'Blue Moon']
print("Blue Moon Transactions:", blueMoonDF.shape[0])
print("Blue Moon Total Scoops:", blueMoonDF['Scoops'].sum())
print("Blue Moon Average Scoops:", round(blueMoonDF['Scoops'].mean(), 2))

print("Conditional Subsetting: Container")

# Waffle Cone
waffleDF = icecreamDF[icecreamDF['Container'] == 'WaffleCone']
print("Waffle Cone Transactions:", waffleDF.shape[0])
print("Waffle Cone Total Scoops:", waffleDF['Scoops'].sum())
print("Waffle Cone Average Scoops:", round(waffleDF['Scoops'].mean(), 2))

# Sugar Cone
sugarDF = icecreamDF[icecreamDF['Container'] == 'SugarCone']
print("Sugar Cone Transactions:", sugarDF.shape[0])
print("Sugar Cone Total Scoops:", sugarDF['Scoops'].sum())
print("Sugar Cone Average Scoops:", round(sugarDF['Scoops'].mean(), 2))

# Groupings
print("Groupings by Flavor:")
print("Transactions by Flavor:")
print(icecreamDF.groupby('Flavor').size())

print("Total Scoops by Flavor:")
print(icecreamDF.groupby('Flavor')['Scoops'].sum())

print("Average Scoops by Flavor:")
print(icecreamDF.groupby('Flavor')['Scoops'].mean().round(2))

print("Groupings by Container:")
print("Transactions by Container:")
print(icecreamDF.groupby('Container').size())

print("Total Scoops by Container:")
print(icecreamDF.groupby('Container')['Scoops'].sum())

print("Average Scoops by Container:")
print(icecreamDF.groupby('Container')['Scoops'].mean().round(2))

```

Shell

```
>>> %Run agg_rand_icdata.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 14/agg_icecream_exercise/code
```

Ice Cream Data:

	Transaction	Flavor	Container	Scoops
0	1	Strawberry	Cup	3
1	2	Strawberry	Cup	1
2	3	Strawberry	SugarCone	3
3	4	Vanilla	WaffleCone	2
4	5	Strawberry	SugarCone	3
..	...	...	...	...
95	96	Blue Moon	WaffleCone	3
96	97	Vanilla	Cup	1
97	98	Strawberry	SugarCone	2
98	99	Strawberry	Cup	2
99	100	Chocolate	Cup	1

[100 rows x 4 columns]

Totals for All Transactions:

Total Scoops (All Transactions): 194

Average Scoop Size (All Transactions): 1.94

Conditional Subsetting: Flavor

Chocolate Transactions: 28

Chocolate Total Scoops: 55

Chocolate Average Scoops: 1.96

Vanilla Transactions: 31

Vanilla Total Scoops: 61

Vanilla Average Scoops: 1.97

Blue Moon Transactions: 16

Blue Moon Total Scoops: 31

Blue Moon Average Scoops: 1.94

Conditional Subsetting: Container

Waffle Cone Transactions: 30

Waffle Cone Total Scoops: 59

Waffle Cone Average Scoops: 1.97

Sugar Cone Transactions: 34

Sugar Cone Total Scoops: 71

Sugar Cone Average Scoops: 2.09

Groupings by Flavor:

Transactions by Flavor:

Flavor

Blue Moon 16

Chocolate 28

Strawberry 25

Vanilla 31

dtype: int64

Total Scoops by Flavor:

Flavor

Blue Moon 31

```

Chocolate  55
Strawberry 47
Vanilla    61
Name: Scoops, dtype: int64
Average Scoops by Flavor:
Flavor
Blue Moon  1.94
Chocolate  1.96
Strawberry 1.88
Vanilla    1.97
Name: Scoops, dtype: float64
Groupings by Container:
Transactions by Container:
Container
Cup        36
SugarCone  34
WaffleCone 30
dtype: int64
Total Scoops by Container:
Container
Cup        64
SugarCone  71
WaffleCone 59
Name: Scoops, dtype: int64
Average Scoops by Container:
Container
Cup        1.78
SugarCone  2.09
WaffleCone 1.97
Name: Scoops, dtype: float64
>>>

```

Transaction, Flavor, Container, Scoops

```

1, Strawberry, Cup, 3
2, Strawberry, Cup, 1
3, Strawberry, SugarCone, 3
4, Vanilla, WaffleCone, 2
5, Strawberry, SugarCone, 3
6, Vanilla, SugarCone, 3
7, Blue Moon, WaffleCone, 1
8, Vanilla, Cup, 3
9, Blue Moon, SugarCone, 3
10, Blue Moon, Cup, 1
11, Blue Moon, Cup, 3
12, Chocolate, SugarCone, 3
13, Strawberry, Cup, 3
14, Vanilla, WaffleCone, 3
15, Vanilla, SugarCone, 3
16, Strawberry, SugarCone, 2
17, Chocolate, WaffleCone, 1
18, Chocolate, Cup, 1
19, Vanilla, WaffleCone, 1
20, Chocolate, SugarCone, 3
21, Vanilla, WaffleCone, 2
22, Vanilla, SugarCone, 2
23, Strawberry, WaffleCone, 1
24, Vanilla, Cup, 1

```

25,Chocolate,SugarCone,1
26,Blue Moon,SugarCone,1
27,Strawberry,Cup,1
28,Strawberry,Cup,3
29,Blue Moon,WaffleCone,3
30,Blue Moon,SugarCone,2
31,Vanilla,SugarCone,2
32,Chocolate,Cup,1
33,Strawberry,WaffleCone,1
34,Blue Moon,SugarCone,3
35,Strawberry,SugarCone,2
36,Strawberry,SugarCone,1
37,Strawberry,Cup,1
38,Chocolate,Cup,1
39,Vanilla,Cup,1
40,Chocolate,WaffleCone,3
41,Chocolate,Cup,1
42,Vanilla,Cup,2
43,Strawberry,Cup,2
44,Vanilla,Cup,3
45,Vanilla,WaffleCone,2
46,Vanilla,Cup,1
47,Vanilla,SugarCone,2
48,Vanilla,SugarCone,3
49,Strawberry,SugarCone,1
50,Chocolate,WaffleCone,1
51,Vanilla,WaffleCone,2
52,Vanilla,Cup,1
53,Vanilla,SugarCone,2
54,Chocolate,WaffleCone,2
55,Chocolate,SugarCone,1
56,Blue Moon,Cup,1
57,Chocolate,SugarCone,3
58,Vanilla,WaffleCone,1
59,Blue Moon,Cup,1
60,Strawberry,Cup,2
61,Chocolate,WaffleCone,1
62,Chocolate,SugarCone,2
63,Chocolate,WaffleCone,3
64,Chocolate,WaffleCone,3
65,Blue Moon,Cup,2
66,Chocolate,SugarCone,3
67,Vanilla,SugarCone,3
68,Strawberry,Cup,3
69,Strawberry,SugarCone,2
70,Chocolate,WaffleCone,2
71,Vanilla,WaffleCone,2
72,Chocolate,SugarCone,3
73,Vanilla,Cup,1
74,Chocolate,WaffleCone,3
75,Chocolate,Cup,2
76,Vanilla,WaffleCone,2
77,Chocolate,WaffleCone,3
78,Vanilla,WaffleCone,1
79,Strawberry,SugarCone,1
80,Blue Moon,Cup,3
81,Strawberry,WaffleCone,2
82,Blue Moon,SugarCone,1
83,Blue Moon,SugarCone,1
84,Strawberry,SugarCone,1

85,Chocolate,WaffleCone,2
86,Strawberry,WaffleCone,2
87,Chocolate,SugarCone,2
88,Vanilla,Cup,2
89,Strawberry,Cup,2
90,Chocolate,SugarCone,1
91,Vanilla,Cup,2
92,Vanilla,Cup,3
93,Vanilla,WaffleCone,2
94,Blue Moon,Cup,2
95,Chocolate,WaffleCone,2
96,Blue Moon,WaffleCone,3
97,Vanilla,Cup,1
98,Strawberry,SugarCone,2
99,Strawberry,Cup,2
100,Chocolate,Cup,1